

Technical Document #814: Blockchain - Comprehensive Overview

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Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Cross-chain technology enables communication between different blockchains. It is essential for interoperability in the blockchain ecosystem.

Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain. They are essential for maintaining network integrity.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions.
- Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain.
- Smart contracts are self-executing contracts with terms directly written into code.